

Objective

A master thesis is an academic paper intended to demonstrate that the author is capable of independently investigating a scientific question in a methodologically correct manner.

Therefore, your work must be

- based on current academic knowledge
- reproducible (correctly cite all literature and data sources)
- well written (clear and coherent structure, stringent argumentation, no errors and typos)
- independently written (no plagiarism; using advice from supervisors but not having them as co-authors)

Timeline

Here is a tentative outline for the upcoming work:

November	Kick-off meeting
January/February	Submission of Research Proposal
February/March	Follow-up meeting
April	Individual meeting
May-June	Final Presentation
mid-July	Delivery of an unbound preliminary full version (optional)
mid-August	Deadline for thesis submission

Kick-off meetings are meant to get you started with the project. We can talk about the topic and how to get started with the empirical part of the project.

An research proposal to be submitted is a detailed plan of the work that I can expect you to produce at the end. It should be 3-5 pages long. Once submitted, I will send you a written feedback over email and we will talk about improvements during the following meeting.

Evaluation

It is my intention to stick to the timeline outlined above. In case you would require more assistance from me, you can ask for additional help over email or through extra individual meetings, but be aware that, in order to get a high grade, you are expected to demonstrate an independence in thinking and an ability to self-organize your learning process. Please keep that in mind.

Language

Never underestimate the importance of clear and concise writing! Convey your message as clearly and simple as possible. Try to keep your sentence short and avoid repetition. In general, your writing style should be objective. That means, you should avoid personal views. If personal views are necessary, they should be identifiable as such.

It goes without saying that impeccable orthography, grammar, and proper punctuation are essential. A good reference is [The Chicago Manual of Style](#).

Structure

The ideal structure of your thesis depends on your research question and your chosen approach to answer this question. However, you should *roughly* follow the following steps:

1. Introduce and motivate your research question
 - This section should answer the question “*Why should we care?*”
2. Provide a synthesis of the current (academic) literature
 - This section should answer the question “*What do we already know about this issue/ topic?*”
3. Explain the data and the chosen statistical method to analyse the research question
 - This section should answer the question “*How did you get to your findings?*”
 - Ideally, a person not involved in your research project (e.g., your thesis supervisor) should be able to obtain the same results by following the steps outlined in this section. In other words, be as precise as possible to ensure replicability. As a useful mnemonic, think of a cake recipe: list all the ingredients and necessary quantities, explain where you obtained them, give clear instructions on how to combine the ingredients, etc.
 - Do not underestimate the amount of data work! Data collection, cleaning and preparation requires a lot of time and some unforeseen difficulties usually arise in every empirical project. Therefore, start with your data work as soon as possible and do not postpone the empirical part to the final months before the deadline.
 - It is generally a good idea to provide summary statistics for the main variables of interest; this will make it easier for the reader to understand the nature of your data and to evaluate the appropriateness of your empirical approach.
4. Present your empirical results in a clear and consistent way
 - This section should answer the question “*What are your findings?*”

- The use of meaningful figures and tables is highly recommended (see below).

5. Interpret and discuss your results

- This section should answer the question “*What do we learn from your results?*”

6. Finally, briefly (!), conclude

As a general advice, try to organize and structure your text by the means of nicely connected paragraphs. Ideally, each paragraph conveys one main point. Importantly, avoid too long paragraphs. As a personal rule of thumb, we generally try to keep the number of lines per paragraph between (approximately) ten to fifteen, however, some paragraphs obviously have to be shorter. Start by drafting a paragraph and convey the point you want to make. If the paragraph turns out to be too long, you can think about how to shorten it.

Read relevant papers related to your topics and try to find a similar structure (sections, subsections). Avoid lengthy textbook-style discussion of basic concepts.

Workload and number of pages:

- Rough guide: The ECTS points for the master thesis correspond to a workload of approximately 55 days
- However, an excellent thesis might require more work
- Pages number is not of first-order importance! Quality comes first.
- Rough guide: 50 pages including bibliography but excluding appendices.
- If you have many and huge charts in your thesis, or you use generous formatting of tables, even 60 pages could be too little content

How to work with literature?

A great source to look for relevant literature is either Google Scholar (or Web of Science). Either use keywords or look for articles that reference the key articles mentioned in your suggested literature list. A good reference point for how influential or important the article is, is the number of citations in Google Scholar. Try to limit your literature list to the papers published in refereed journals for academics and practitioners.

Recent and forthcoming articles in high-reputation finance journals include, e.g., JOF, RFS, JFE, RoF, JFQA; eventually econ and field journals. Concerning working papers, think about quality. Typically high quality when presented at a top conference (e.g., AFA, WFA, EFA, Cavalcade, FIRS) or from well-respected authors.

Referencing books is appropriate to a certain extend. Try to make sure that your academic literature list is not heavily populated by books. The relative

share of books in references should definitely not exceed 30-40%. Why - (text)books mostly present digested information that once appeared in the form of a scientific article. One of the things you want to learn while writing this thesis is how to work with original sources of information. That is, scientific articles.

I would strongly encourage you to include a section in your thesis that discusses relevance of the topic today. To get a sense of what journalists and non-professionals think about this issue, check articles published by Economist.com. Wall Street Journal is a bit more specialized source, but also a good platform to find opinions on what is important today and why.

Presentation of (empirical) results

Your empirical results often represent the centerpiece of your thesis and the one part that required the most amount of work. Therefore, make sure that you present your results in the best possible way. Specifically, ensure that your results are easily readable:

- Avoid small axis labels and overlaps in your figures
- Use an interpretable scale for the axis labels in your figures
- Tables and figures should always include a caption that briefly describes all variables/data shown

As a rule of thumb, the reader should be able to understand your tables and figures from reading the captions, without reading the text. We strongly urge you to look at published academic papers (e.g., in The Journal of Finance, The Review of Financial Studies, The Journal of Financial Economics) for examples on how to design figures and structure tables. This is particularly important for regression outputs (horizontal lines, centered columns, consistent number of decimals points, etc.)

Software for statistical analysis

In order to conduct analysis for your master thesis, you must be familiar with one of the software for statistical analysis. Excel is not enough!!! I strongly recommend R or STATA.

- R: a free statistical package. You can find some basic tips on how to use R here: [click here for R](#),
- STATA: I recommend a textbook: *Microeconometrics with STATA* by A. Colin Cameron and Pravin K. Trivedi and you can find it in the WU library.

Data

Important market data sources are: Bloomberg, CRSP (access through WRDS), Compustat (access through WRDS). Data on financial characteristics of banks in Europe are available in Bankscope. For the USA, please see Compustat North America file. Data on market indexes as well as their composition can be obtained from CRSP and Bloomberg. A complete list of available data providers can be found [here](#).

Editor and References

I require you to stick to the referencing style that is adopted in the finance academic literature. An example of a recently published paper, where you can see how the authors reference previous work, can be found here: [Example](#).